

MECH 223 – ENGINEERING STATICS

PRACTICE EXAMPLES

- 1) A weather balloon is tethered at point $A = (0, 12.0, 0)^m$ above the origin by three cables attached to ground anchors B, C, and D on the horizontal ground plane ($y = 0^m$):
- Anchor B: Located at $(-4.00, 0, 0)^m$ (along the negative x-axis).
 - Anchor C: Located at $(3.00, 0, 5.00)^m$.
 - Anchor D: Located at $(0, 0, -6.00)^m$ (along the negative z-axis).

Knowing that the tension in cable AD is $T_{AD} = 900.0^N$, determine:

- The tensions in cables AB and AC (T_{AB} and T_{AC})
- The total vertical buoyant lift force P exerted by the balloon at point A.

Solution:

❖ Position Vectors and Cable lengths:

Finding the vectors pointing from tether node $A(0, 12.0, 0)^m$ to each anchor point:

- Vector AB:

$$\vec{AB} = (-4.00 - 0)\hat{i} + (0 - 12.0)\hat{j} + (0 - 0)\hat{k} = -4.00\hat{i} - 12.0\hat{j}^m$$

$$|\vec{AB}| = \sqrt{(-4.00)^2 + (-12.0)^2} = 12.65^m$$
- Vector AC:

$$\vec{AC} = (3.00 - 0)\hat{i} + (0 - 12.0)\hat{j} + (5.00 - 0)\hat{k} = 3.00\hat{i} - 12.0\hat{j} + 5.00\hat{k}^m$$

$$|\vec{AC}| = \sqrt{(3.00)^2 + (-12.0)^2 + (5.00)^2} = 13.34^m$$
- Vector AD:

$$\vec{AD} = (0 - 0)\hat{i} + (0 - 12.0)\hat{j} + (-6.00 - 0)\hat{k} = -12.0\hat{j} - 6.00\hat{k}^m$$

$$|\vec{AD}| = \sqrt{(-12.0)^2 + (-6.00)^2} = 13.42^m$$

❖ Unit Vectors & Force Vector Representations

Dividing each position vector by its length to get direction unit vectors ($\hat{\lambda} = \frac{\vec{r}}{|\vec{r}|}$):

Unit Vectors:

$$\hat{\lambda}_{AB} = \frac{-4.00}{12.65}\hat{i} - \frac{12.0}{12.65}\hat{j} = -0.316\hat{i} - 0.949\hat{j}$$

$$\hat{\lambda}_{AC} = \frac{3.00}{13.34}\hat{i} - \frac{12.0}{13.34}\hat{j} + \frac{5.00}{13.34}\hat{k} = 0.225\hat{i} - 0.899\hat{j} + 0.375\hat{k}$$

$$\hat{\lambda}_{AD} = \frac{-12.00}{13.42}\hat{j} - \frac{6.00}{13.42}\hat{k} = -0.894\hat{j} - 0.447\hat{k}$$

The vertical balloon lift force acts straight up along the +y axis:

$$P = P\hat{j}$$

❖ Static Equilibrium Equations ($\sum \mathbf{F} = \mathbf{0}$)

Setting the sum of components in all three coordinate directions to zero:

- Horizontal x-axis ($\sum F_x = 0$):

$$-0.316T_{AB} + 0.225T_{AC} = 0 \rightarrow T_{AC} = \left(\frac{0.316}{0.225}\right)T_{AB} = 1.406T_{AB}$$

- Transverse z-axis ($\sum F_z = 0$):

$$0.375T_{AC} - 0.447T_{AD} = 0$$

- Vertical y-axis ($\sum F_y = 0$):

$$-0.949T_{AB} - 0.899T_{AC} - 0.894T_{AD} + P = 0$$

$$P = 0.949T_{AB} + 0.899T_{AC} + 0.894T_{AD}$$

❖ **Cable Tension and Lift Evaluations:**

Given $T_{AD} = 900.0^N$:

Evaluating for T_{AC} from $\sum F_z = 0$:

$$0.375T_{AC} = 0.447 * 900.0 = 402.5$$

$$T_{AC} = \frac{402.48}{0.375} = 1074^N$$

Evaluating for T_{AB} from $\sum F_x = 0$:

$$1074 = 1.406T_{AB} \rightarrow T_{AB} = 763.8^N$$

Evaluating for vertical lift P from $\sum F_y = 0$:

$$P = 0.949 * 763.8 + 0.899 * 1074 + 0.895 * 900 = 2495^N$$

2) A vertical communications mast is fixed at the ground origin $O(0,0,0)^m$ and supported at a top collar $A(0,10.0,0)^m$ by three guy cables anchored to ground supports at points B, C, and D ($y = 0^m$):

- Anchor B: Located at $(-5.00,0,0)^m$ (along the negative x-axis)
- Anchor C: Located at $(4.00,0,6.00)^m$
- Anchor D: Located at $(0,0,-5.00)^m$ (along the negative z-axis)

A downward load and wind gust create an equivalent upward axial support reaction $P = P\hat{j}$ applied at top node A. Knowing that the maximum allowable tension capacity in cable AD is $T_{AD} = 1100^N$, determine:

- 1- The required tension in cables AB and AC
- 2- The required magnitude of reaction force P to maintain static equilibrium.

Solution:

❖ **Coordinates and Position Vectors**

Finding the direction vectors pointing from top collar $A(0,10.0,0)^m$ down to each ground anchor:

- Vector AB:

$$\vec{AB} = (-5.00 - 0)\hat{i} + (0 - 10.0)\hat{j} + (0 - 0)\hat{k} = -5.00\hat{i} - 10.0\hat{j}^m$$

$$|\vec{AB}| = \sqrt{(-5.00)^2 + (-10.0)^2} = 11.18^m$$

- Vector AC:

$$\vec{AC} = (4.00 - 0)\hat{i} + (0 - 10.0)\hat{j} + (6.00 - 0)\hat{k} = 4.00\hat{i} - 10.0\hat{j} + 6.00\hat{k}^m$$

$$|\vec{AC}| = \sqrt{(4.00)^2 + (-10.0)^2 + (6.00)^2} = 12.33^m$$

- Vector AD:

$$\vec{AD} = (0 - 0)\hat{i} + (0 - 10.0)\hat{j} + (-5.00 - 0)\hat{k} = -10.0\hat{j} - 5.00\hat{k}^m$$

$$|\vec{AD}| = \sqrt{(-10.0)^2 + (-5.00)^2} = 11.18^m$$

❖ **Unit Vectors & Force Vector Representations**

Dividing each vector by its length to determine direction unit vectors ($\hat{\lambda} = \frac{\vec{r}}{|\vec{r}|}$):

$$\hat{\lambda}_{AB} = \frac{-5.00}{11.18}\hat{i} - \frac{10.0}{11.18}\hat{j} = -0.447\hat{i} - 0.894\hat{j}$$

$$\hat{\lambda}_{AC} = \frac{4.00}{12.33}\hat{i} - \frac{10.0}{12.33}\hat{j} + \frac{6.00}{12.33}\hat{k} = 0.324\hat{i} - 0.811\hat{j} + 0.487\hat{k}$$

$$\hat{\lambda}_{AD} = \frac{-10.0}{11.18}\hat{j} - \frac{5.00}{11.18}\hat{k} = -0.894\hat{j} - 0.447\hat{k}$$

❖ **Static Equilibrium Equations ($\sum \mathbf{F} = \mathbf{0}$):**

Summing the components along each principal coordinate axis to zero:

- I. Horizontal x-axis ($\sum F_x = 0$):

$$-0.447T_{AB} + 0.324T_{AC} = 0 \rightarrow T_{AC} = 1.379T_{AB}$$

- II. Transverse z-axis ($\sum F_z = 0$):

$$0.487T_{AC} - 0.447T_{AD} = 0$$

III. Vertical y-axis ($\sum F_y = 0$):

$$-0.894T_{AB} - 0.811T_{AC} - 0.894T_{AD} + P = 0$$

$$\rightarrow P = 0.894T_{AB} + 0.811T_{AC} + 0.894T_{AD}$$

❖ Tension and Support Reaction Calculations

Given $T_{AD} = 1100^N$:

- Evaluating for T_{AC} from $\sum F_z = 0$:

$$0.4867T_{AC} = 0.447 * 1100 = 492.92 \rightarrow T_{AC} = 1010.7^N$$

- Evaluating for T_{AB} from $\sum F_x = 0$:

$$1010.7 = 1.379T_{AB} \rightarrow T_{AB} = 733.2^N$$

- Evaluating for vertical reaction force P from $\sum F_y = 0$:

$$P = 0.894 * (733.2) + 0.811 * (1010.7) + 0.894 * (1100) = 2459.4^N$$

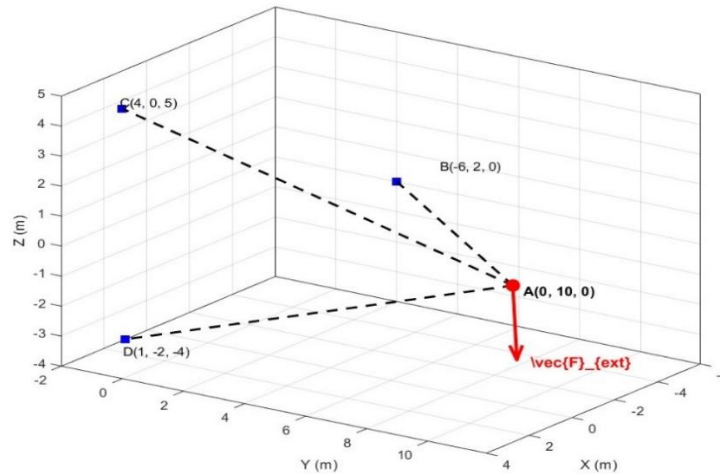
3) During an orbital assembly mission, a central structural connector node at position $A(0,10.0,0)^m$ is held in position relative to a space station module by three high-strength tether cables attached to anchor brackets on the module's hull at B, C, and D:

- Anchor B: Located at $(-6.00,2.00,0)^m$
- Anchor C: Located at $(4.00,0,5.00)^m$
- Anchor D: Located at $(1.00,-2.00,-4.00)^m$

During a thruster maneuver, a satellite tether applies an external pulling force F_{ext} to node A directed along the vector $u_{ext} = 4\hat{i} + 3\hat{j} - 2\hat{k}$.

If the maximum tension capacity any individual tether cable can support before snapping is $T_{max} = 1600^N$, determine:

- 1- Which tether cable reaches its maximum tension capacity first.
- 2- The maximum allowable magnitude the thruster pulling force F_{ext} .



Solution:

❖ Position Vectors and Cable Lengths

Calculating the position vectors pointing from the central node $A(0,10.0,0)^m$ to each anchor point:

- Vector AB:

$$\vec{AB} = (-6.00 - 0)\hat{i} + (2.00 - 10.0)\hat{j} + (0 - 0)\hat{k} = -6.00\hat{i} - 8.00\hat{j}^m$$

$$|\vec{AB}| = \sqrt{(-6.00)^2 + (-8.00)^2} = 10.0^m$$

- Vector AC:

$$\vec{AC} = (4.00 - 0)\hat{i} + (0 - 10.0)\hat{j} + (5.00 - 0)\hat{k} = 4.00\hat{i} - 10.0\hat{j} + 5.00\hat{k}^m$$

$$|\vec{AC}| = \sqrt{(4.00)^2 + (-10.0)^2 + (5.00)^2} = 11.87^m$$

- Vector AD:

$$\vec{AD} = (1.00 - 0)\hat{i} + (-2.00 - 10.0)\hat{j} + (-4.00 - 0)\hat{k} = 1.00\hat{i} - 12.0\hat{j} - 4.00\hat{k}^m$$

$$|\vec{AD}| = \sqrt{(1.00)^2 + (-12.0)^2 + (-4.00)^2} = 12.69^m$$

❖ Unit Vectors & Force Representations

Dividing each vector by its length to determine direction unit vectors ($\hat{\lambda} = \frac{\vec{r}}{|\vec{r}|}$):

$$\hat{\lambda}_{AB} = \frac{-6.0}{10}\hat{i} - \frac{8.0}{10}\hat{j} = -0.60\hat{i} - 0.80\hat{j}$$

$$\hat{\lambda}_{AC} = \frac{4.00}{11.87}\hat{i} - \frac{10.0}{11.87}\hat{j} + \frac{5.00}{11.87}\hat{k} = 0.337\hat{i} - 0.842\hat{j} + 0.421\hat{k}$$

$$\hat{\lambda}_{AD} = \frac{1.00}{12.69}\hat{i} - \frac{12.00}{12.69}\hat{j} - \frac{4.00}{12.69}\hat{k} = 0.079\hat{i} - 0.946\hat{j} - 0.315\hat{k}$$

- External force unit vector ($\hat{\lambda}_{ext}$):

$$|\vec{u}_{ext}| = \sqrt{4^2 + 3^2 + (-2)^2} = 5.385^m$$

$$\hat{\lambda}_{ext} = \frac{4}{5.385}\hat{i} + \frac{3}{5.385}\hat{j} - \frac{2}{5.385}\hat{k} = 0.743\hat{i} + 0.557\hat{j} - 0.371\hat{k}$$

❖ **Static Equilibrium Equations ($\sum \vec{F} = \vec{0}$):**

Summing components along all three Cartesian axes:

- I. Horizontal x-axis ($F_x = 0$):

$$-6.00T_{AB} + 0.337T_{AC} + 0.079T_{AD} + 0.743F_{ext} = 0 \quad [Equation 1]$$

- II. Vertical y-axis ($\sum F_y = 0$):

$$-0.800T_{AB} - 0.842T_{AC} - 0.946T_{AD} + 0.557F_{ext} = 0 \quad [Equation 2]$$

- III. Transverse z-axis ($\sum F_z = 0$):

$$0.421T_{AC} - 0.315T_{AD} - 0.371F_{ext} = 0 \quad [Equation 3]$$

❖ **Linear System Solution**

- Expressing T_{AC} in terms of T_{AD} and F_{ext} from [Equation 3]:

$$T_{AC} = \frac{0.315T_{AD} + 0.371F_{ext}}{0.421} = 0.749T_{AD} + 0.882F_{ext} \quad [Equation 4]$$

- Substituting [Equation 4] into [Equation 1] to express T_{AB} :

$$0.6T_{AB} = 0.337(0.749T_{AD} + 0.882F_{ext}) + 0.079T_{AD} + 0.743F_{ext}$$

$$0.06T_{AB} = 0.331T_{AD} + 1.04F_{ext}$$

$$\rightarrow T_{AB} = 0.552T_{AD} + 1.733F_{ext} \quad [Equation 5]$$

- Substituting [Equation 4] and [Equation 5] into [Equation 2] to isolate T_{AD} :

$$-0.8(0.552T_{AD} + 1.733F_{ext}) - 0.842(0.749T_{AD} + 0.882F_{ext}) - 0.945T_{AD} + 0.557F_{ext} = 0$$

$$\rightarrow (-0.441 - 0.630 - 0.946)T_{AD} + (-1.387 - 0.743 + 0.557)F_{ext} = 0$$

$$\rightarrow -2.018T_{AD} - 1.572F_{ext} = 0 \rightarrow T_{AD} = -0.779F_{ext}$$

- Physical Interpretation Note: The mathematical negative sign indicates that a pulling force along direction vector $u_{ext} = 4\hat{i} + 3\hat{j} - 2\hat{k}$ puts cable AD into compression (which cables cannot support, meaning cable AD goes slack: $T_{AD} = 0$).

Setting $T_{AD} = 0$ in the remaining active cable balance system ([Equation 1] and [Equation 3]):

- From [Equation 3]:

$$0.421T_{AC} = 0.371F_{ext} \rightarrow T_{AC} = 0.882F_{ext}$$

- From [Equation 1]:

$$0.6T_{AB} = 0.337(0.882F_{ext}) + 0.743F_{ext} = 1.04F_{ext} \rightarrow T_{AB} = 1.733F_{ext}$$

❖ **Critical Tether Identification & Maximum Force**

1- Limiting Cable:

Comparing the proportional load factors ($T_{AB} = 1.733F_{ext}$ vs. $T_{AC} = 0.882F_{ext}$), Tether AB carries the largest proportion of the external load and will reach maximum capacity first.

2- Maximum Allowable Force F_{ext} :

Setting $T_{AB} = T_{max} = 1600^N$:

$$1600 = 1.733F_{ext} \rightarrow F_{ext} = 923.16^N$$

Checking remaining active cable:

$$T_{AC} = 0.882 * 923.16 = 814.23^N < 1600^N \text{ (Safe)}$$

- 4) An offshore mooring support node located at $A(0,12,0)^m$ is anchored to the seabed ($y = 0^m$) by three mooring cables attached at points B, C, and D:
- Anchor B: Located at $(-8.00,0,0)^m$ (along the negative x-axis)
 - Anchor C: Located at $(5.00,0,4.00)^m$
 - Anchor D: Located at $(0,0,-9.00)^m$ (along the negative z-axis)

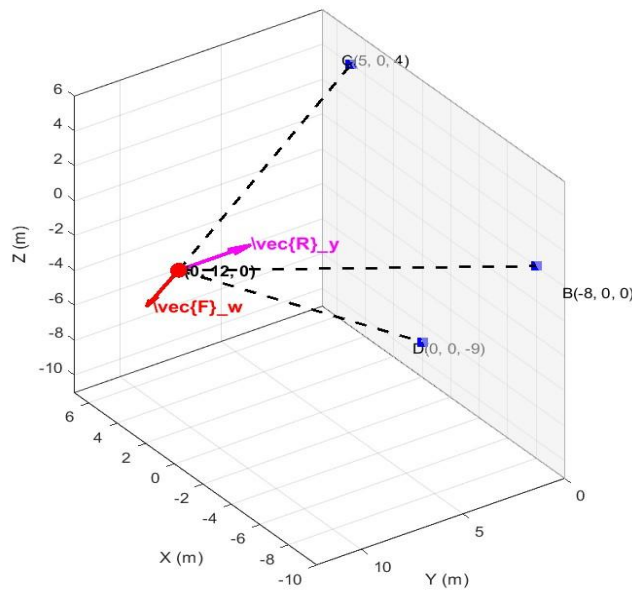
A ocean current creates a horizontal wave drag force at node A:

$$F_w = 2.00\hat{i} - 3.00\hat{k}^{kN}$$

The central spud column provides a downward axial thrust force at A acting along the negative y-axis ($R_y = -R_y\hat{j}$). Each mooring cable has a maximum safe operating tension capacity of $T_{max} = 20.0^{kN}$.

Knowing that cable AD carries a measured tension of $T_{AD} = 6.00^{kN}$:

- 1- Evaluate the tensions in cables AB and AC (T_{AB}, T_{AC}) and the required downward thrust R_y .
- 2- Determine which cable is critical and evaluate system safety under $T_{max} = 20.0^{kN}$
- 3- Calculate the Maximum Safe Load Factor ($k = \frac{T_{max}}{T_{critical}}$) and the required percentage reduction in wave drag force for safe operation.



Solution

❖ Position Vectors & Cable Lengths

Finding vectors pointing from node $A(0,12,0)^m$ to each anchor point:

- Vector AB:

$$\vec{AB} = (-8.00 - 0)\hat{i} + (0 - 12.0)\hat{j} + (0 - 0)\hat{k} = -8.00\hat{i} - 12.00\hat{j}^m$$

$$|\vec{AB}| = \sqrt{(-8.00)^2 + (-12.00)^2} = 14.42^m$$

- Vector AC:

$$\vec{AC} = (5.00 - 0)\hat{i} + (0 - 12.0)\hat{j} + (4.00 - 0)\hat{k} = 5.00\hat{i} - 12.0\hat{j} + 4.00\hat{k}^m$$

$$|\vec{AC}| = \sqrt{(5.00)^2 + (-12.00)^2 + (4.00)^2} = 13.60^m$$

- Vector AD:

$$\vec{AD} = (0 - 0)\hat{i} + (0 - 12.0)\hat{j} + (-9.00 - 0)\hat{k} = -12.0\hat{j} - 9.00\hat{k}^m$$

$$|\vec{AD}| = \sqrt{(-12.00)^2 + (-9.00)^2} = 15.0^m$$

❖ Unit Vectors

Dividing position vectors by their respective lengths:

$$\hat{\lambda}_{AB} = \frac{-8.00}{14.42}\hat{i} - \frac{12.0}{14.42}\hat{j} = -0.555\hat{i} - 0.832\hat{j}$$

$$\hat{\lambda}_{AC} = \frac{5.00}{13.60}\hat{i} - \frac{12.0}{13.60}\hat{j} + \frac{4.00}{13.60}\hat{k} = 0.368\hat{i} - 0.882\hat{j} + 0.294\hat{k}$$

$$\hat{\lambda}_{AD} = \frac{-12.00}{15.00}\hat{j} - \frac{9.00}{15.00}\hat{k} = -0.80\hat{j} - 0.60\hat{k}$$

❖ Static Equilibrium Equations ($\sum F = 0$)

Summing forces along coordinate axes including $F_w = 2.00\hat{i} - 3.00\hat{k}^{kN}$:

- I) Horizontal x-axis ($\sum F_x = 0$):

$$-0.555T_{AB} + 0.368T_{AC} + 2.00 = 0$$

- II) Transverse z-axis ($\sum F_z = 0$):

$$0.294T_{AC} - 0.60T_{AD} - 3.00 = 0$$

- III) Vertical y-axis ($\sum F_y = 0$):

$$-0.832T_{AB} - 0.882T_{AC} - 0.80T_{AD} - R_y = 0$$

❖ Tension & Structural Load Calculations

Given $T_{AD} = 6.00^{kN}$:

- Evaluating T_{AC} from $\sum F_z = 0$:

$$0.294T_{AC} - 0.60(6.00) - 3.00 = 0$$

$$\rightarrow T_{AC} = 22.44^{kN}$$

- Evaluating T_{AB} from $\sum F_x = 0$:

$$-0.555T_{AB} + 0.368(22.44) + 2.00 = 0$$

$$\rightarrow T_{AB} = 18.48^{kN}$$

- Evaluation downward reaction force R_y from $\sum F_y = 0$:

$$R_y = 0.832(18.48) + 0.882(22.44) + 0.80(6.00)$$

$$\rightarrow R_y = 39.98^{kN}$$

❖ Safety Evaluation & Load Reduction

- Critical Member: Cable AC carries the highest tension ($T_{AC} = 22.44^{kN}$).
- Safety Status: Since $T_{AC} = 22.44^{kN} > T_{max} = 20.0^{kN}$, the capacity limit is exceeded (unsafe).
- Maximum Safe Load Factor (k):

$$k = \frac{T_{max}}{T_{AC}} = \frac{20.0^{kN}}{22.44^{kN}} = 0.89$$

- Required Drag Force Reduction Percentage:

$$Reduction = (1 - k) * 100 = (1 - 0.89) * 100 = 10.88\%$$